SKY-X

Autonomous AI-Powered Search and Rescue Drone

1. Project Title

SKY-X – Autonomous AI-Powered Search and Rescue Drone

2. Problem Statement

During disasters such as earthquakes, building collapses, floods, fires, and industrial accidents, search-and-rescue teams may have to enter dangerous areas with poor visibility and uncertain information about victim locations. Manual searching can take time and may expose rescuers to unnecessary risks. A system is needed that can quickly survey affected areas, identify possible victims, assess their priority, and provide useful information to rescue operators.

3. Solution Proposed

SKY-X is an autonomous search-and-rescue drone designed to support rescue teams during emergency situations. It uses RGB and thermal cameras along with distance sensing to collect information from the environment. An onboard edge-AI system processes the sensor data to detect possible victims and estimate their priority. ROS 2 and a PX4-based flight controller support autonomous navigation, while the mission can be replanned when an important detection is found. A ground station provides operators with live video, telemetry, victim locations, priority levels, alerts, and search coverage.

Key Features

• RGB and thermal sensing for victim detection
• Distance sensing for obstacle awareness
• Edge AI for onboard detection and assessment
• Dynamic victim priority scoring
• Autonomous navigation using ROS 2 and PX4
• Dynamic mission replanning
• Ground-station monitoring
• Operator override, geofencing and failsafe controls

Technology Stack

Layer	Technology / Mechanism
Perception	RGB camera + thermal camera + distance sensing
AI processing	Onboard edge AI computer
Robotics middleware	ROS 2
Flight control	PX4-based flight controller
Autonomy	Obstacle detection/avoidance + mission planning and replanning
Operator interface	Ground station with video, telemetry, locations, priorities, alerts and coverage

4. Implementation Plan

The implementation will be carried out in controlled stages so that the AI, sensing, navigation, and safety functions can be tested before real-world deployment. The first stage will define the complete system architecture, mission workflow, sensing arrangement, AI pipeline, flight-control interfaces, and operator interface. The second stage will focus on simulation, where autonomous search, victim detection, prioritization, navigation, and mission replanning can be tested safely. The third stage will integrate RGB and thermal detection with sensor-fusion logic on the onboard edge computer. The fourth stage will connect ROS 2, PX4, sensing, and autonomy functions and test obstacle avoidance and mission execution. The fifth stage will develop the ground station to display video, telemetry, victim locations, priority levels, alerts, and search coverage. Finally, controlled validation tests will be performed with increasing scenario complexity to evaluate system performance and safety behavior.

The main beneficiaries are search-and-rescue teams, emergency-response operators, and disaster-management organizations. The expected outputs include a simulation-validated autonomous search workflow, an integrated RGB/thermal victim-detection and sensor-fusion pipeline, victim priority scoring and dynamic mission-replanning logic, ROS 2 and PX4 integration, a ground-station interface, and controlled-test results.

5. Safety and Efficiency Analysis

Safety is a primary requirement because the drone will operate in emergency environments where autonomous flight can affect the drone and nearby people. SKY-X therefore uses a dedicated safety layer in addition to the AI detection system. Operator override allows a human operator to intervene whenever autonomous behavior becomes inappropriate or unsafe. Geofencing keeps the drone within the defined operating area, while failsafe mechanisms provide predefined responses to selected faults or abnormal conditions. Distance sensing supports obstacle detection and avoidance to reduce collision risk. Simulation-first validation allows autonomous behavior to be tested before controlled real-world deployment, and human oversight keeps rescue operators informed through live telemetry, video, alerts, and victim information. These measures can also improve efficiency by reducing search time, combining complementary RGB and thermal information, prioritizing important detections, reducing rescuer exposure to hazardous areas, and maintaining continuous situational awareness through the ground station.

6. Scalability and Future Development

The SKY-X architecture is designed to remain modular so that new sensors, AI models, mission behaviors, and operator interfaces can be added without redesigning the complete platform. Future development can extend the system from a single drone to coordinated multi-drone teams for covering larger disaster areas. Additional environmental or ranging sensors can be integrated when required, while improved AI models can provide better confidence estimation, condition assessment, and victim prioritization. Richer 3D mapping can improve localization and situational awareness in complex structures. Communication-resilience methods can also be developed for areas with limited or intermittent connectivity. The mission-planning system can further be adapted for floods, earthquakes, industrial incidents, wildfires, and other emergency scenarios. Keeping sensing, AI inference, mission planning, flight control, and operator monitoring as modular components will make the platform easier to adapt to new environments and applications.

7. Innovation Advantage

The main innovation of SKY-X is that it goes beyond simply detecting a person. The system combines multimodal sensing, AI-based victim assessment, dynamic priority scoring, and autonomous mission replanning. RGB and thermal sensing provide complementary information when normal visual detection may be affected by darkness or smoke. After detecting a possible victim, the system can estimate priority using detection confidence, thermal evidence, estimated severity, and proximity. This information can then influence the search mission so that higher-priority detections can be investigated first. The ground station also brings together video, telemetry, victim locations, priorities, alerts, and search coverage in one interface. In this way, SKY-X connects perception to action: it detects people, estimates their priority, uses that information to guide the search mission, and reports the results to a human rescue operator.

8. Visual Impact and User Journey

Before – Conventional Response: Disaster area → rescuers manually search difficult or hazardous areas → limited visibility and uncertain victim location → time-consuming search and greater exposure to hazards.

After – With SKY-X: Drone surveys the disaster area → RGB, thermal and distance sensing → AI detects, assesses and prioritizes victims → autonomous navigation and dynamic mission replanning → ground station provides location, priority, alerts and live video → rescue team acts with improved situational awareness.

User Journey: The rescue operator first launches or assigns the SKY-X search mission in the defined emergency area. The operator then monitors live video and telemetry from the ground station and receives detected-person locations and dynamic priority levels. When a high-priority detection is identified, SKY-X can investigate it and replan the mission. The operator uses the reported location, priority, and visual information to support rescue decisions, while search-coverage information shows which areas have already been examined. This changes the operator's role from manually searching for uncertain targets to using structured and prioritized information from an autonomous aerial platform.

Conclusion

SKY-X proposes an autonomous aerial approach to search and rescue that combines computer vision, thermal sensing, distance sensing, edge AI, ROS 2, and PX4. By detecting and prioritizing possible victims, adapting the search mission, and providing information to rescue operators, the system aims to make emergency search faster, safer, and more informed. The simulation-first and controlled-validation approach also provides a practical path toward developing and testing the proposed system.